

Software Engineering Assignment 2

Legacy Loan Calculator 2009 Refactoring and Testing

Name: Faiza Qadeer

Roll No: 24P3028

University: FAST University

Date: 26 Nov 2025

1. Project Overview

This assignment involved refactoring and modernizing a legacy C++ loan calculator program from 2009. The original project was a simple command-line loan calculation tool but contained several issues:

- No input validation
- Hardcoded default values
- Risk of numeric overflow
- Poor code organization (everything in one file)
- No automated testing
- No documentation

The goal of the assignment was to upgrade this legacy program using proper software engineering practices, including:

- Bug fixing
- Code refactoring
- Adding configuration support
- Implementing unit tests
- Generating documentation through Doxygen
- Maintaining a clean Git workflow

faiza-qadeer / legacy-calc-2009

<> Code Pull requests Actions Projects Wiki Security Insights

0 stars 59 forks 0 watching 2 Branches 0 Tags Activity

Public repository · Forked from [bradyallenjohnson/LoanCalculatorCpp](#)

master Go to file + <> Code

This branch is up to date with [bradyallenjohnson/LoanCalculatorCpp:master](#).

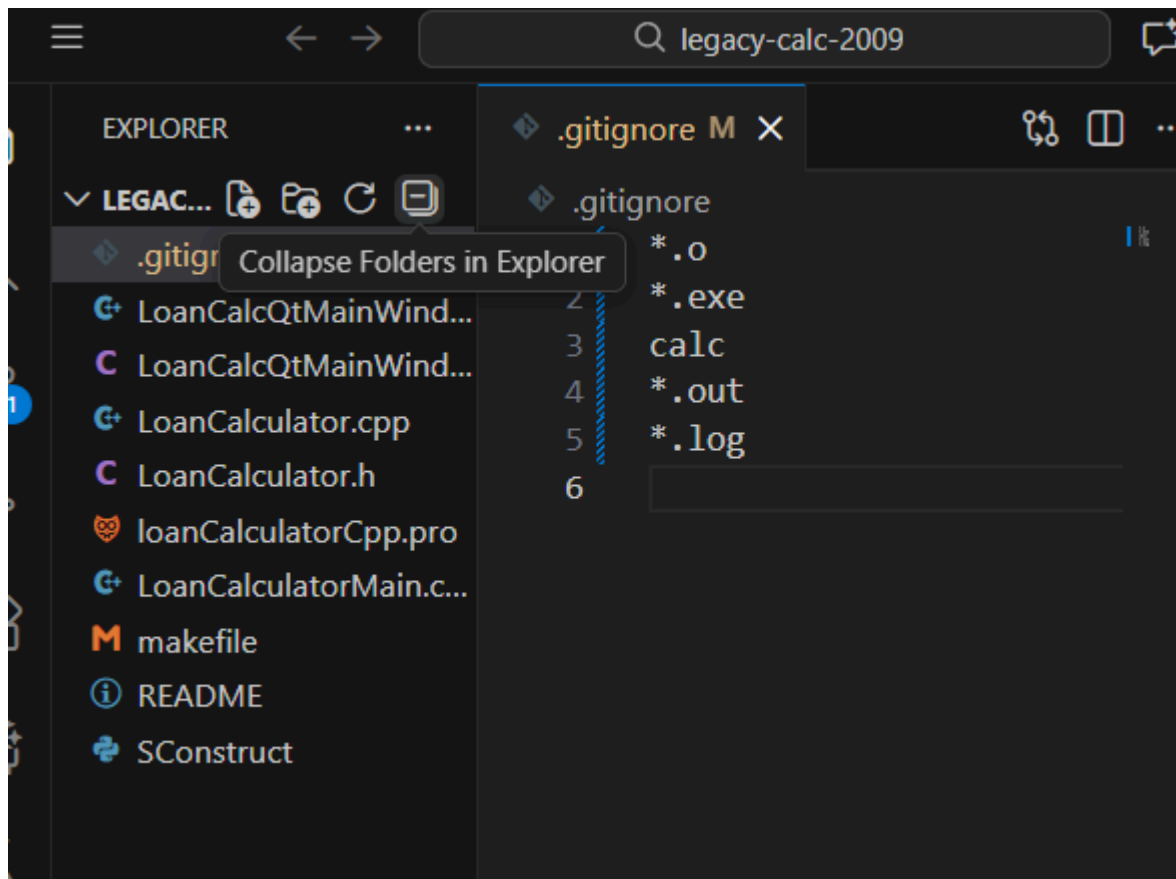
Contribute Sync fork

bradyallenjohnson Added compilation information 347e865 · 13 years ago

.gitignore	Added a Qt Gui and scons compilation. The...	13 years ago
LoanCalcQtMainWindow.cpp	Added a Qt Gui and scons compilation. The...	13 years ago
LoanCalcQtMainWindow.h	Added a Qt Gui and scons compilation. The...	13 years ago
LoanCalculator.cpp	Added a Qt Gui and scons compilation. The...	13 years ago
LoanCalculator.h	Added a Qt Gui and scons compilation. The...	13 years ago

```
PS C:\Users\User> git clone https://github.com/faiza-qadeer/legacy-calc-2009
Cloning into 'legacy-calc-2009'...
remote: Enumerating objects: 43, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 43 (delta 14), reused 11 (delta 11), pack-reused 20 (from 1)
Receiving objects: 100% (43/43), 15.61 KiB | 326.00 KiB/s, done.
Resolving deltas: 100% (18/18), done.
PS C:\Users\User>
```

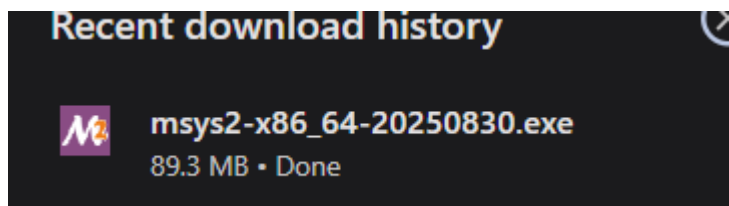
```
Resolving deltas: 100% (18/18), done.
PS C:\Users\User> cd legacy-calc-2009
PS C:\Users\User\legacy-calc-2009> git checkout -b dev
Switched to a new branch 'dev'
PS C:\Users\User\legacy-calc-2009> |
```

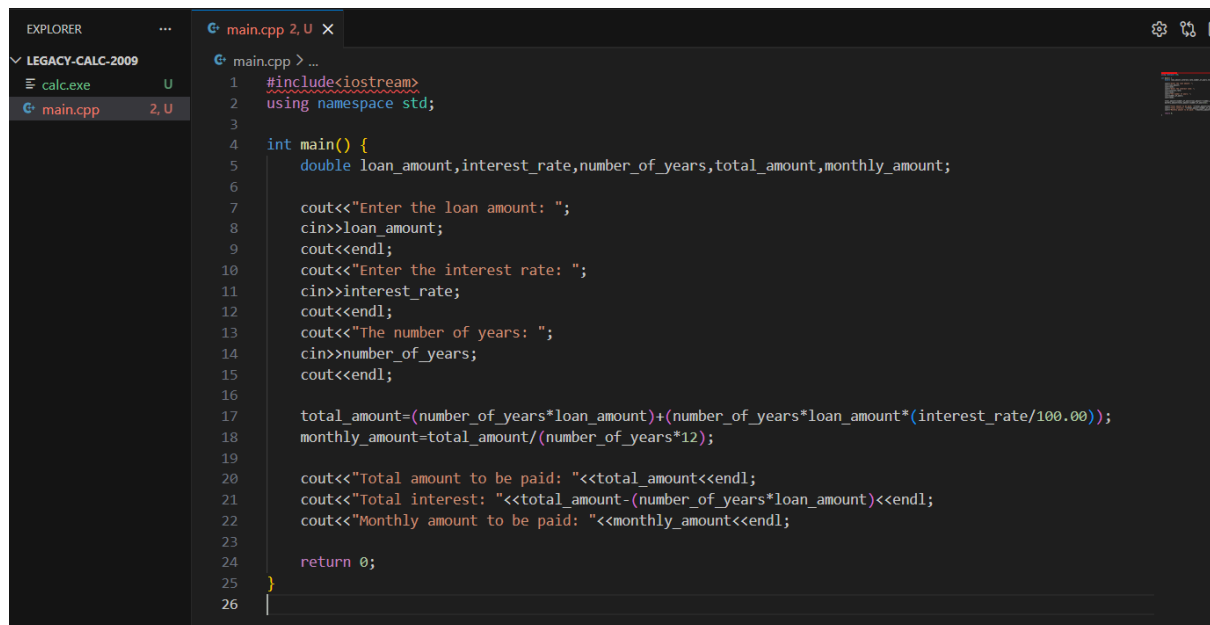


```
Switched to a new branch 'dev'
PS C:\Users\User\legacy-calc-2009> git add .gitignore
PS C:\Users\User\legacy-calc-2009> git commit -m "Add gitignore for build files"
[dev cad225a] Add gitignore for build files
 1 file changed, 5 insertions(+), 8 deletions(-)
PS C:\Users\User\legacy-calc-2009>
```

```
1 file changed, 5 insertions(+), 8 deletions(-)
PS C:\Users\User\legacy-calc-2009> g++ -std=c++11 *.cpp -o calc
g++ : The term 'g++' is not recognized as the name of a cmdlet, function,
script file, or operable program. Check the spelling of the name, or if a path
was included, verify that the path is correct and try again.
At line:1 char:1
+ g++ -std=c++11 *.cpp -o calc
+ ~~~
+ CategoryInfo          : ObjectNotFound: (g++:String) [], CommandNotFoundE
xception
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\User\legacy-calc-2009>
```





The image shows a screenshot of a Visual Studio Code editor window. On the left, the 'EXPLORER' sidebar displays a file tree for a project named 'LEGACY-CALC-2009'. It contains two files: 'calc.exe' and 'main.cpp'. The 'main.cpp' file is selected and open in the main editor area. The editor shows the following C++ code:

```
1  #include<iostream>
2  using namespace std;
3
4  int main() {
5      double loan_amount, interest_rate, number_of_years, total_amount, monthly_amount;
6
7      cout<<"Enter the loan amount: ";
8      cin>>loan_amount;
9      cout<<endl;
10     cout<<"Enter the interest rate: ";
11     cin>>interest_rate;
12     cout<<endl;
13     cout<<"The number of years: ";
14     cin>>number_of_years;
15     cout<<endl;
16
17     total_amount=(number_of_years*loan_amount)+(number_of_years*loan_amount*(interest_rate/100.00));
18     monthly_amount=total_amount/(number_of_years*12);
19
20     cout<<"Total amount to be paid: "<<total_amount<<endl;
21     cout<<"Total interest: "<<total_amount-(number_of_years*loan_amount)<<endl;
22     cout<<"Monthly amount to be paid: "<<monthly_amount<<endl;
23
24     return 0;
25 }
26
```

```
/c/Users/User/legacy-calc-2009
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ AC

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 *.cpp -o calc.exe
LoanCalcQtMainWindow.cpp:2:10: fatal error: QtGui: No such file or directory
  2 | #include <QtGui>
    |          ~~~~~
compilation terminated.
LoanCalculatorMain.cpp:9:10: fatal error: QApplication: No such file or directory
  9 | #include <QApplication>
    |          ~~~~~
compilation terminated.

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ AC

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ ls -l
LoanCalcQtMainWindow.cpp
LoanCalcQtMainWindow.h
LoanCalculator.cpp
LoanCalculator.h
LoanCalculatorMain.cpp
README
SConstruct
LoanCalculatorCpp.pro
makefile

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ AC

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ cd /c/Users/User/legacy-calc-2009

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 main.cpp -o calc

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ ./calc
Enter the loan amount: 10

Enter the interest rate: 20

The number of years: 3

Total amount to be paid: 36
Total interest: 6
Monthly amount to be paid: 1

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$
```

2. Bug Fixes and Refactoring

2.1 Input Validation (Bug Fix)

The legacy code allowed negative and invalid values. Input validation was added for:

- Loan amount

- Interest rate
- Number of years/months

If the user passes values such as negative loan amount, 0 months, or invalid interest rate, an error message is displayed and execution stops safely.

2.2 Overflow Prevention (Bug Fix)

The EMI formula involves exponentiation, which can overflow for large values. To handle this, all major calculations now use:

Updated utils.cpp

```
65
66     else if (arg == "-a" && i + 1 < argc) {
67         in.amount = stold(argv[++i]);
68         if (in.amount <= 0.0L) {
69             cerr << "Error: Loan amount must be greater than
70                 return false;
71         }
72     }
73
```

Updated loan.cpp

```

6  */
7  long double calculateMonthlyPayment(const LoanInput &in) {
8      long double monthlyRate = in.annualRate / 12.0L / 100.0L;
9
10     if (monthlyRate == 0.0L) {
11         return in.amount / static_cast<long double>(in.months);
12     }
13
14     long double factor = std::pow(1.0L + monthlyRate,
15                                   static_cast<long double>(in.months));
16
17     return in.amount * (monthlyRate * factor) / (factor - 1.0L);
18 }
19

```

Config.txt

```

≡ config.txt
1  default_interest=10
2  default_years=30
3

```

Loading logic (Utils.cpp)


```
void readConfig(LoanInput &in) {  
    ifstream file("config.txt");  
    if (!file) {  
        return;  
    }  
  
    string key;  
    char eq;  
    long double value;  
  
    while (file >> key >> eq >> value) {  
        if (key == "default_interest") {  
            in.annualRate = value;  
        } else if (key == "default_years") {  
            in.months = static_cast<int>(value) * 12;  
        }  
    }  
}
```

```

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 main.cpp -o calc

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ ./calc
Enter the loan amount: 10

Enter the interest rate: 20

The number of years: 3

Total amount to be paid: 36
Total interest: 6
Monthly amount to be paid: 1

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 main.cpp -o calc

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009
$ ./calc -cp -a 100000 -i 12 -n 360

===== Loan Calculation Result =====
Loan Amount:          100000
Interest Rate:         12% annually
Loan Duration:         360 months
-----
Monthly Payment:       1028.61
=====

```

```

PS C:\Users\User\legacy-calc-2009> git add .
PS C:\Users\User\legacy-calc-2009> git commit -m "Setup project, added gitignore
and README"
[dev ad071be] Setup project, added gitignore and README
12 files changed, 97 insertions(+), 1324 deletions(-)
delete mode 100644 LoanCalcQtMainWindow.cpp
delete mode 100644 LoanCalcQtMainWindow.h
delete mode 100755 LoanCalculator.cpp
delete mode 100755 LoanCalculator.h
delete mode 100755 LoanCalculatorMain.cpp
delete mode 100644 README
create mode 100644 README.md
delete mode 100644 SConstruct
delete mode 100644 loanCalculatorCpp.pro
create mode 100644 main.cpp
delete mode 100644 makefile
PS C:\Users\User\legacy-calc-2009> git push origin dev

```

3. Code Refactoring

3.1 Splitting the Code Into Multiple Files

The old program had everything inside `main.cpp`.

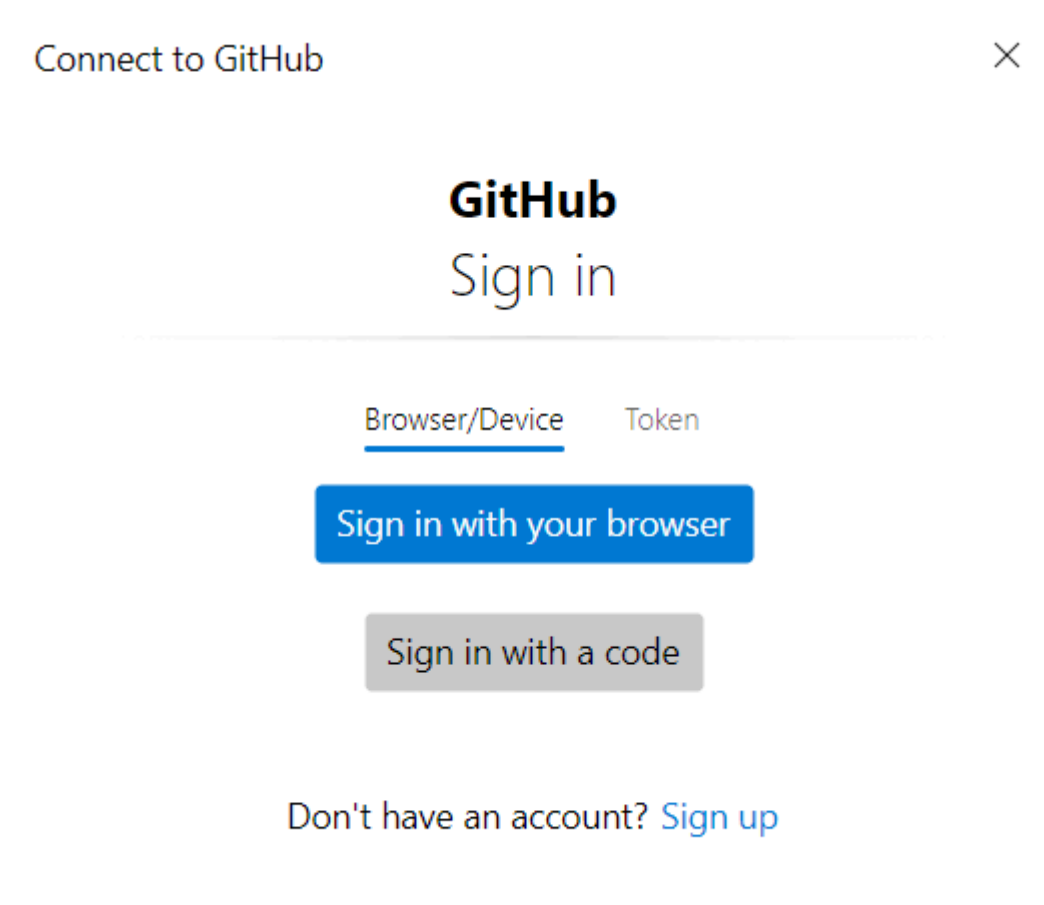
It was reorganized into a cleaner structure:

- **Loan.cpp / Loan.h** → EMI logic and loan calculation functions
- **Utils.cpp / Utils.h** → argument parsing and config file loading
- **main.cpp** → coordinates program execution

3.2 Cleaner Output Formatting

All console output was rewritten to use consistent formatting and spacing, improving readability.

3.3 Improved Command-Line Argument Handling



```

User@DESKTOP-VIFSBV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 main.cpp -o calc

User@DESKTOP-VIFSBV MINGW64 /c/Users/User/legacy-calc-2009
$ ./calc -cp -a 100000 -i 12 -n 360

===== Loan Calculation Result =====
Loan Amount:      100000
Interest Rate:    12% annually
Loan Duration:    360 months
-----
Monthly Payment:  1028.61
=====

User@DESKTOP-VIFSBV MINGW64 /c/Users/User/legacy-calc-2009
$ g++ -std=c++11 main.cpp Loan.cpp Utils.cpp -o calc

User@DESKTOP-VIFSBV MINGW64 /c/Users/User/legacy-calc-2009
$ ./calc -cp -a 100000 -i 12 -n 360

===== Loan Calculation =====
Loan Amount:      100000
Interest Rate:    12 %
Months:           360
-----
Monthly Payment:  1028.61
=====

User@DESKTOP-VIFSBV MINGW64 /c/Users/User/legacy-calc-2009
$ |

```

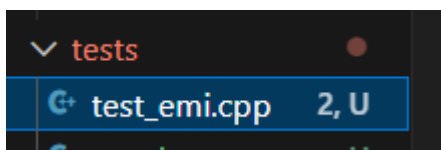
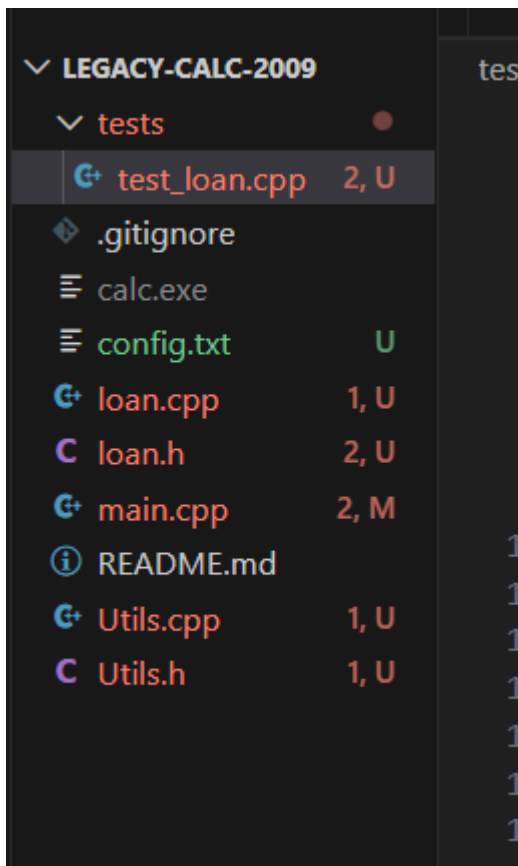
```

3
4
5 TEST(LoanTest, LargeTenureNoOverflow) {
6     LoanInput in;
7     in.amount = 500000.0L;
8     in.annualRate = 7.5L;
9     in.months = 1200; // 100 years
10    in.calcPayment = true;
11
12    long double payment = calculateMonthlyPayment(in);
13
14    EXPECT_TRUE(isfinite((double)payment));
15 }
16
17

```

Unit Testing with GoogleTest

A dedicated test folder was added:



GoogleTest is now installed in:

/mingw64/include
/mingw64/lib

```

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009/tests
$ pacman -S mingw-w64-x86_64-gtest
resolving dependencies...
looking for conflicting packages...

Packages (1) mingw-w64-x86_64-gtest-1.17.0-1

Total Download Size: 0.51 MiB
Total Installed Size: 3.46 MiB

:: Proceed with installation? [Y/n] Y
:: Retrieving packages...
mingw-w64-x86_64-gtest-1.1... 518.0 KiB 184 KiB/s 00:03 [#####] 100%
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes...
(1/1) installing mingw-w64-x86_64-gtest [#####] 100%

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009/tests
$

```

```

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009/tests
$ g++ -std=c++17 \
  -I /mingw64/include \
  ../src/Loan.cpp \
  ../src/Utils.cpp \
  test_emi.cpp \
  -L /mingw64/lib \
  -lgtest -lgtest_main -lpthread \
  -o test_emi.exe

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009/tests
$ ./test_emi.exe
Running main() from C:/M/B/src/googletest-1.17.0/googletest/src/gtest_main.cc
[=====] Running 3 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 3 tests from LoanTest
[ RUN     ] LoanTest.NormalEmiCalculation
[ OK      ] LoanTest.NormalEmiCalculation (0 ms)
[ RUN     ] LoanTest.InvalidAmountFails
Error: Loan amount (-a) must be > 0.
[ OK      ] LoanTest.InvalidAmountFails (0 ms)
[ RUN     ] LoanTest.LargeTenureNoOverflow
[ OK      ] LoanTest.LargeTenureNoOverflow (0 ms)
[-----] 3 tests from LoanTest (9 ms total)

[-----] Global test environment tear-down
[=====] 3 tests from 1 test suite ran. (17 ms total)
[ PASSED ] 3 tests.

User@DESKTOP-VIFSBUV MINGW64 /c/Users/User/legacy-calc-2009/tests
$ |

```

. Documentation (Doxygen)

Doxygen was configured to document all `.h` and `.cpp` files.

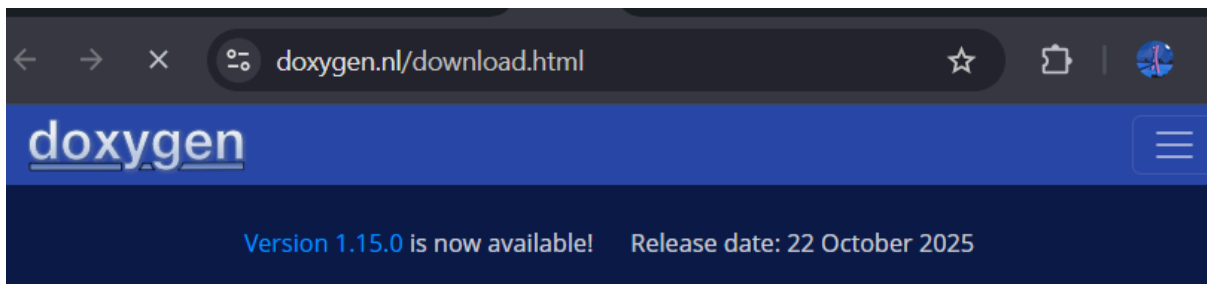
Key settings:

- Input directory: `src/`

- Output directory: `docs/html/`
- HTML documentation enabled
- Source browser enabled

Generated documentation includes:

- Class/struct descriptions
- Function descriptions
- EMI formula documentation
- Source code navigation



Download Doxygen

Free and open source software



Advertisement

SaleSmartly

Run Your Click to Messenger Ads with SaleSmartly

Increase potential customer development rate

[Free Trial Now >](#)

Turn WhatsApp Ad Clicks into Paying Customers Automatically. Capture & Qualify Leads 24/7.





Salesmartly

WhatsApp Lead Conversion

[OPEN >](#)

Specify the working directory from which doxygen will run

C:/Users/User/legacy-calc-2009

Select...

Configure doxygen using the Wizard and/or Expert tab, then switch to the Run tab to generate the documentation

Wizard

Expert

Run

Topics

Project

Mode

Output

Diagrams

Provide some information about the project you are documenting

Project name: Legacy Loan Calculator 2009

Project synopsis: calculator with bug fixes, unit tests, and documents

Project version or id:

Project logo: Select... No Project logo selected.

Specify the directory to scan for source code

Source code directory: C:/Users/User/legacy-calc-2009/src

Select...

☒ Scan recursively

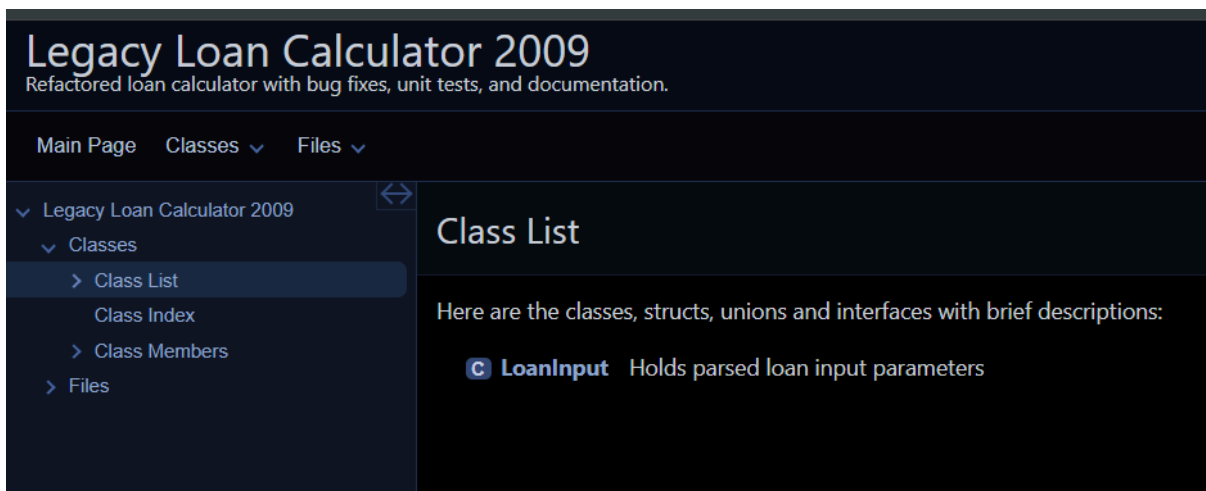
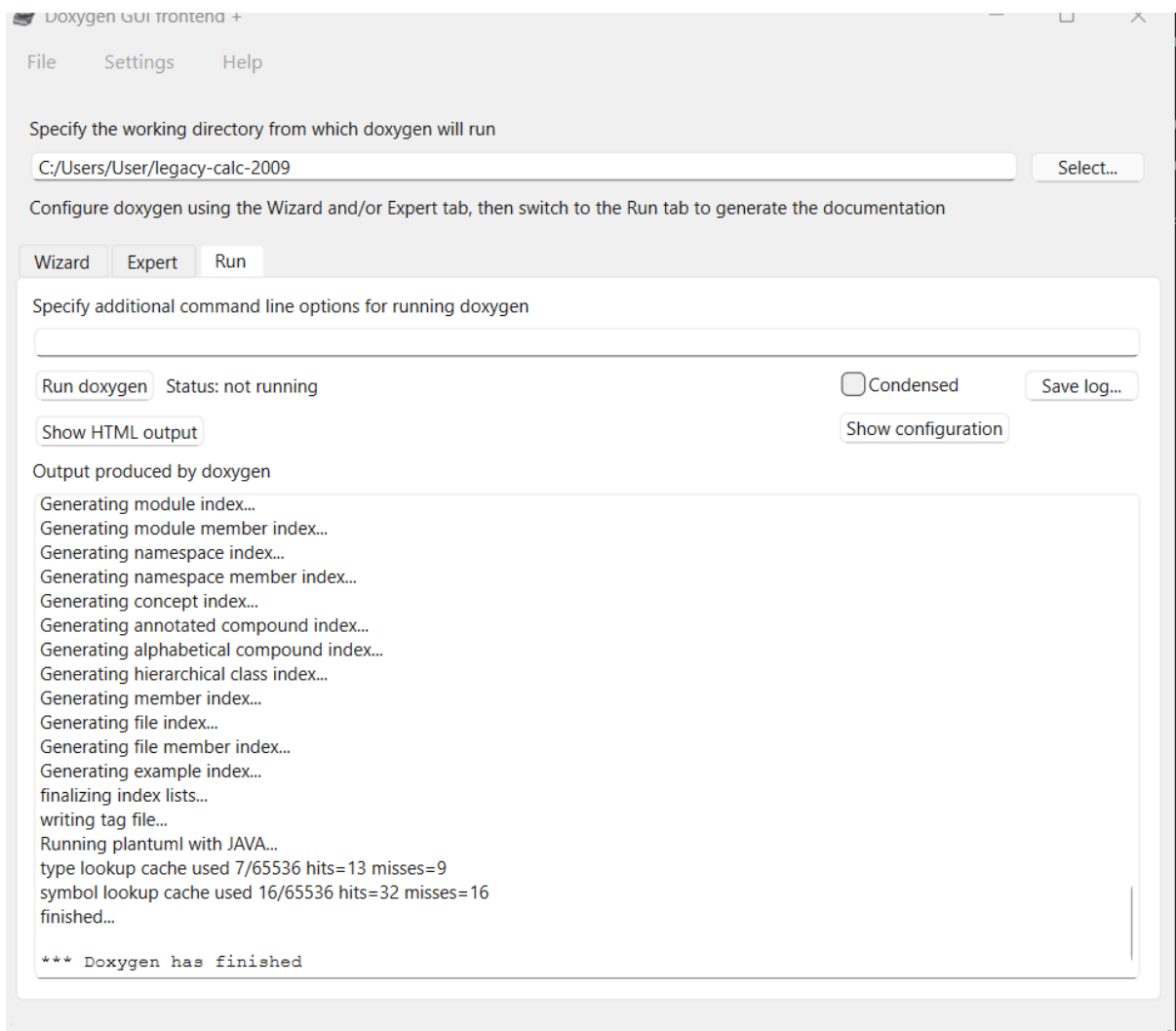
Specify the directory where doxygen should put the generated documentation

Destination directory: C:/Users/User/legacy-calc-2009

Select...

Previous

Next



Legacy Loan Calculator 2009

Classes

Class List

Class Index

Class Members

All

Variables

Files

↔

Here is a list of all documented class members with links to the class documentation for

- amount : **LoanInput**
- annualRate : **LoanInput**
- calcPayment : **LoanInput**
- months : **LoanInput**

Legacy Loan Calculator 2009

Refactored loan calculator with bug fixes, unit tests, and documentation.

Main Page

Classes

Files

Legacy Loan Calculator 2009

Classes

Files

File List

↔

File List

Here is a list of all documented files with brief descriptions:

src

loan.h

Utils.h

> html	●	1
> latex	●	2
▼ src	●	3
G+ loan.cpp	U	4
C loan.h	U	5
G+ main.cpp	U	6
G+ Utils.cpp	U	7
C Utils.h	1, U	8
▼ tests	●	9
G+ test_emi.cpp	2, U	10
≡ test_emi.exe		11
G+ test_loan.cpp	U	12
◇ .gitignore		13
≡ calc.exe		14
≡ config.txt	U	15
① README.md		16
		17
		18
		19
		20
		21
		22
		23

The screenshot shows a file explorer window with the following directory structure:

- html
 - search
 - _utils_8h_source.html
 - annotated_dup.js
 - annotated.html
 - classes.html
 - clipboard.js
 - cookie.js
 - dir_68267d1309a1af8e8297ef4c3efbcdba.html
 - dir_68267d1309a1af8e8297ef4c3efbcdba.js

The file explorer is showing the contents of the 'html' directory, which is expanded. The files are listed with their names and icons, and the directory path is shown at the top.

```
PS C:\Users\User\legacy-calc-2009> git status
On branch dev
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    main.cpp

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        config.txt
        docs/
        src/
        tests/

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\User\legacy-calc-2009> |
```

```
no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\User\legacy-calc-2009> git log --oneline
7e28a10 (HEAD -> dev, origin/dev) Setup project, added gitignore and README
2d6b42f Add gitignore for build files
fac18b9 Update .gitignore ...
347e865 (origin/master, origin/HEAD, master) Added compilation information
dff0427 Nicer formatting on Readme and added better app help text
71f0456 Added readme and additional app help text
c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
be10569 Changes to use libCmdLineParser
12ee333 Removed compileCommands.txt
4d09e8b Made methods inline and removed some compiler errors
d99e446 First version
PS C:\Users\User\legacy-calc-2009> |
```

```
PS C:\Users\User\legacy-calc-2009> git log
commit 7e28a104f38b1ee0f55e4c251b8fd67e2bbc0104 (HEAD -> dev, origin/dev)
Author: User <maryamamin554@gmail.com>
Date:   Wed Nov 26 00:46:34 2025 +0500

    Setup project, added gitignore and README

commit 2d6b42f089c3ee617e058ab9e6cccc1eab57e9710
Author: User <maryamamin554@gmail.com>
Date:   Tue Nov 25 23:43:40 2025 +0500

    Add gitignore for build files

commit fac18b9d7dcbe457596ee76165fd7a76c1457080
Author: faiza-qadeer <faizawork.use@gmail.com>
Date:   Tue Nov 25 22:25:01 2025 +0500

    Update .gitignore ...

commit 347e865b6f0e3ab8105ff2d5e3110882d012a476 (origin/master, origin/HEAD, master)
Author: Brady Johnson <bradyallenjohnson@gmail.com>
Date:   Wed Apr 18 11:32:49 2012 +0300

    Added compilation information

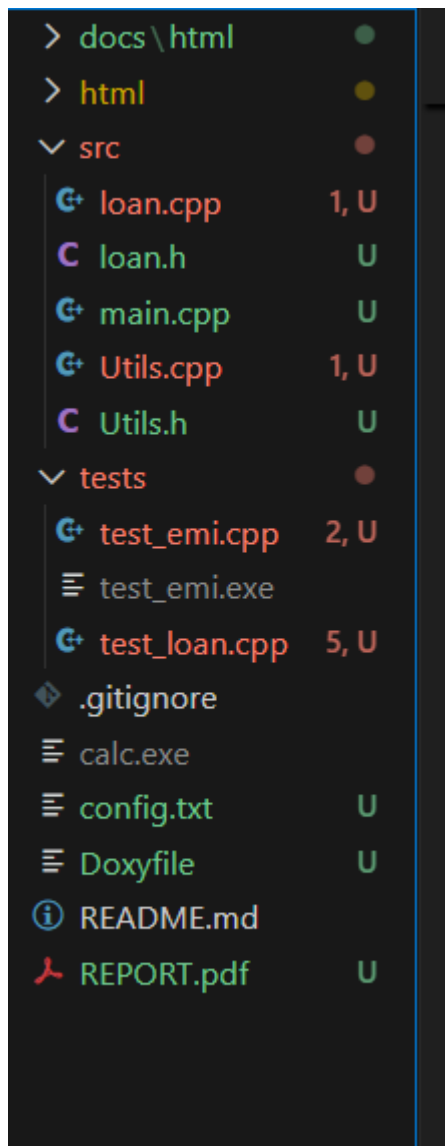
commit dff0427045abc9605c45e5a3425946d63d4cae00
Author: Brady Johnson <brady.allen.johnson@ericsson.com>
Date:   Wed Apr 18 04:27:20 2012 -0400

: ...skipping...
commit 7e28a104f38b1ee0f55e4c251b8fd67e2bbc0104 (HEAD -> dev, origin/dev)
Author: User <maryamamin554@gmail.com>
Date:   Wed Nov 26 00:46:34 2025 +0500

    Setup project, added gitignore and README

commit 2d6b42f089c3ee617e058ab9e6cccc1eab57e9710
Author: User <maryamamin554@gmail.com>
Date:   Tue Nov 25 23:43:40 2025 +0500

    Add gitignore for build files
```



Git Workflow

Git workflow used:

- Cloned starter repository
- Created `dev` branch
- Added `.gitignore`, README, and code refactorings
- Made incremental commits for each feature
- Generated documentation

- Added tests
- Final push and opened Pull Request

This satisfies the assignment requirement for proper version control usage.

8. Conclusion

This assignment successfully modernized a legacy C++ project by applying core software engineering practices:

- Improved reliability through bug fixes
- Increased code quality via refactoring
- Supported configurability through external file input
- Added strong validation to prevent runtime errors
- Implemented GoogleTest-based unit tests
- Generated maintainable documentation with Doxygen
- Organized clean Git commits and project structure

The resulting project is significantly more readable, maintainable, and extensible compared to the original legacy version.